



Automated classification of acute leukemia on a heterogeneous dataset using machine learning and deep learning techniques

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ABSTRACT

Today, artificial intelligence and deep learning techniques constitute a prominent part in the area of medical sciences. These techniques help doctors detect diseases early and reduce their burden as well as chances of errors. However, experiments based on deep learning techniques require large and well-annotated dataset. This paper introduces a novel dataset of 500 peripheral blood smear images, containing normal, Acute Myeloid Leukemia and Acute Lymphoblastic Leukemia images. The dataset comprises almost 1700 cancerous blood cells. The size of the dataset is increased by adding images of a publicly available dataset and forming a heterogeneous dataset. The heterogeneous dataset is used for the automated binary classification task, which is one of the major tasks of the proposed work. The proposed work perform binary as well as three-class classification tasks involving state-of-the-art techniques based on machine learning and deep learning. For binary classification, the proposed work achieved an accuracy of 97% when fully connected layers along with the last three convolutional layers of VGG16 are fine tuned and 98% for DenseNet121 along with support vector machine. For three-class classification task, an accuracy of 95% is obtained for ResNet50 along with support vector machine. The preparation of the novel dataset is done under the opinion of various expertise that will help the scientific community for medical research supported by machine learning models.

1. Introduction

Among the prominent cause of death, worldwide, cancer ranks second. About 9.6 million people died in 2018 due to cancer. From a report of the world health organization (WHO), about 1 in 6 deaths is due to cancer [1]. Leukemia is one of the types of cancer that affects blood cells. It is basically a heterogeneous clonal irregularity of haemopoietic progenitor cells commonly known as c"lasts" [2]. Blasts do not have the ability to differentiate normally, and they also do not acknowledge general regulators of proliferation. In the absence of proper treatment, this may cause fatal infection, organ infiltration or bleeding. Bone marrow starts producing cancerous blood cells that force out normal blood cells [3,4]. White blood cells (WBC), also known as leucocytes or leukocytes, are more likely to become cancerous than red blood cells (RBC), and platelets.

Two types of aberrant leukocytes can become cancerous: myeloid cells and lymphoid cells. Leukemia caused due to aberrant myeloid cells

is known as myeloid or myelogenous leukemia, and caused due to abnormal lymphoid cells is known as lymphoblastic or lymphocytic leukemia. Depending on how fast cancerous cells grow, leukemia can be categorised as:

1. acute (if growth is fast)
2. chronic (if growth is slow).

So, there are mainly four forms of leukemia:

1. Acute Myeloid Leukemia (AML)
2. Acute Lymphoblastic Leukemia (ALL)
3. Chronic Myeloid Leukemia (CML)
4. Chronic Lymphocytic Leukemia (CLL).

In India, annually more than 10,000 instances of childhood leukemia have been revealed. ALL is the cause of 60% to 85% of all childhood

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Table 1

Comparison of the existing public databases with the proposed dataset for leukemia detection.

Authors	Leukemia type	Size of images	No. of images
Agaian et al. [2]	AML & normal	184 × 138	80
Labati et al. [16]	ALL & normal	2592 × 1944	108
Kumar et al. [20]	B-ALL & MM	2560 × 1920	190
Proposed dataset	ALL, AML & normal	4908 × 3264	500

leukemias while AML accounts for 15% to 20% of childhood leukemias. According to GLOBOCAN 2018 report, it is responsible for more than 0.3 million deaths worldwide. During the same year 32,471 people died in India due to leukemia [5].

Though any age group can be affected, it is most common in adults over 55 and children under 15. Since the precise reason behind leukemia is unknown, so, it is tough to diagnose. A patient suffering from leukemia can present with symptoms like fever, tiredness, weakness, or aches in bones and joints, which are very much identical to typical diseases like flu. If there is such symptoms, then blood tests, such as a full blood count, bone marrow examination, and renal function, should be done [6]. The choice for nature of therapy also differs from radiation therapy, chemotherapy, biological therapy, and bone marrow transplant [7]. The diagnosis of leukemia relies heavily on the operator's experience and fatigue levels. With the help of the computer, image processing techniques is developing very rapidly. This development leads to the automatic detection, classification, and counting of WBCs with the help of microscopic images of blood cells. This computerisation is very important to enhance performance and diminish the load of doctors. Nevertheless, there are some difficulties in obtaining data from WBCs due to factors like varying shapes and sizes of cells, irregular brightness, and different microscopic views [8].

1.1. Related work

Significant endeavours have been dedicated to leukemia image analysis domain and especially, neural networks (NN) as a powerful pixel classifier of medical images into cancerous and non-cancerous categories. In 1996, Kovalev et al. [9] apply statistical approaches along with neural network for WBC image identification. They introduce an efficient and robust technique for separating WBCs from a complicated background. They test their proposed method on a batch of 662 images of five different kinds of WBC. They acquire the images from 22 blood stains under different conditions. The recognition accuracy of 98% is reported. Foran et al. [10] develop an intelligent image repository with the help of a telemicroscopy system. This helps doctors and researchers working at different clinical and research labs to interactively consult and obtain computer-assisted decision support. Their image dataset is formed using 19 cases including 14 lymphoproliferative cases [5 CLL, 5 mantle cell lymphoma (MCL), 4 follicular center cell lymphoma (FCC)], and 5 normal (benign) cases. The image database consists of 261 images (98 CLL, 66 MCL, 38 FCC, and 59 benign cells). Images are taken with the aid of a Leica microscope, 40× planachromatic objective. They achieve success in 83% of the cases studied in discriminating among three lymphoproliferative disorders and healthy cells. Kim et al. [11] form a system to detect the abnormality of WBC and RBC of humans to help pathologists. They use a dyed blood image dataset of two hundred patients. Their dataset consists of 15 classes of RBC, and 5 classes of WBC. To train the neural network classifier, 680 RBC images, and 410 WBC images are used. All the images are captured with a colour CCD camera attached to a microscope with magnification of four hundred times. The dimension of each image is 640 × 480. Principal component analysis (PCA) technique is used to decrease the dimensions of features, which helps in improving the efficiency of the classifier. Leave-one-out method [12] is applied for the testing part.

Piuri et al. [13] propose a method for detection and classification of leucocytes. Firstly, the leucocytes in the blood image are separated from the other blood cells. Morphological indices corresponding to the leucocytes are extracted, and finally, the leucocytes are classified with the help of a neural network classifier as basophil, lymphocyte, eosinophil, monocyte, and neutrophil. For testing, some images are randomly chosen from the image repository [14], which consists of 113 images containing 134 expert-annotated leucocytes. Fabio Scotti [6,15] presents a method to identify ALL cases from microscopic blood-film images. First of all, the leucocytes in the blood image are separated from the others blood cells, then morphological analysis of the leucocytes is done to detect ALL cases. The system's performance is evaluated by randomly selected images from an image repository furnished by the M. Tettamanti Research Center for Childhood Leukemias and Hematological Diseases, Monza, Italy. This database contains 113 images, containing almost 8400 blood cells. Out of 8400, 150 are lymphocytes annotated by experienced oncologists as healthy or blast. To enhance the microscopic images, the undesired background is removed. An accuracy of 92% is reported. In 2011, Labati et al. [16] introduce an image database of blood specimens of leukemic patients and healthy persons. There are 108 images in this publicly available dataset taken with the help of a PowerShot G5 camera mounted on a microscope. Various magnifications varying from 300 to 500 are used to capture the images. Each image is taken in three-channel, i.e., red-green-blue (RGB) colour space. Images are having colour depth of 8 bits per colour channel, i.e., a total of 24-bit colour depth, dimension of 2592 × 1944 and saved in JPG format. The dataset comprises of nearly 39,000 blood elements, where 510 are lymphoblasts. Agaian et al. [2] present an automatically AML detecting methodology in blood microscopic images. They use a dataset of 80 high-quality 184 × 138 size images acquired from the American Society of Haematology (ASH) [17]. 98% accuracy is reported for the lymphoblast detection. Rawat et al. [18] propose a computer assisted classification framework for diagnosis of ALL, AML and normal cases, and the French, American, and British (FAB) characterization of ALL. They use a database consisting of 240 images taken from ASH [17]. The database contains 100 images of healthy subjects, 60 of ALL subjects and 80 of AML subjects. They consider only three sub-types of AML, i.e. M2, M3, and M5 due to database inadequacy. An accuracy of 99.5% for the 3-class classification of acute leukemia (normal, ALL, and AML), and 97.1% and 98.5% for the 3-class classification of subtypes of ALL and AML, respectively. Qin et al. [19] propose a classifier based on deep residual learning theory which they test on 40 types of leukocyte images. An accuracy of 76.84% is reported on the test dataset. Kumar et al. [20] present classification of two types of blood cancer, i.e., ALL, and Multiple Myeloma (MM) with the help of a convolutional neural network (CNN) consisting of 2 convolution layers, 2 max pooling layers, and 5 fully connected layers, including the final output layer. They use a dataset consisting of 90 B-ALL, and 100 MM images and achieved an overall accuracy of 97.2%. Genovese et al. [21] propose a machine learning-based technique based on deep learning for focus quality estimation, adaptive unsharping, and classification of blood samples into ALL and normal categories. They achieved an accuracy of 96.84. Table 1 shows the comparison of the existing public databases of microscopic images with the proposed dataset for leukemia detection. It is observed that the proposed dataset contains both ALL and AML types with the highest number of high resolution images.

1.2. Motivation and contributions

From the literature, it can be inferred that a remarkable amount of work has been done on detection of leukemia. But these works are done on small and homogeneous datasets. Agaian et al. [2] clearly mention that the major hurdle in the improvement of performance of an automated classification system is the paucity of sizeable, annotated, and publicly available database. Huge database with proper annotation is a must requirement for development and validation of machine learning

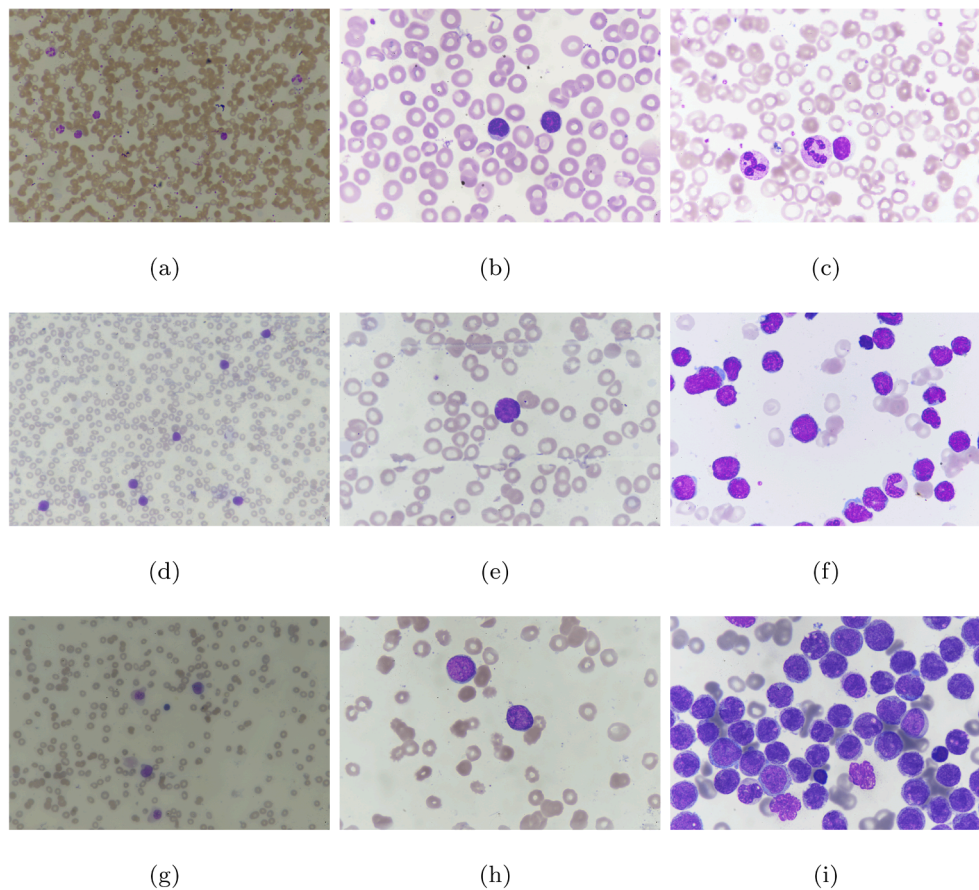


Fig. 1. Samples from the proposed dataset. (a)-(c) show images of normal cases. (d)-(f) show images of ALL. (g)-(i) show images of AML. The first column contains the images taken at 400 $\times$ magnification and the second and third columns contain images taken at 1000 $\times$ magnification.

and deep learning algorithms. This paper addresses the above challenge. The contributions of this article are as follows.

1. A novel dataset consisting of microscopic images of ALL, AML, and normal cases is proposed.
2. A heterogeneous dataset is formed by adding images of a publicly available dataset, i.e., ALL-IDB [16] to the proposed dataset. The heterogeneous dataset is called mixed dataset in this work.
3. Proposed and mixed datasets are used to detect leukemia with the help of machine learning and deep learning techniques. Obtained results are analysed quantitatively and qualitatively.

The organization of the paper is as follows. The proposed dataset is introduced in Section 2. Machine learning and deep learning methods for leukemia detection are discussed in Section 3. Experiments performed and the results are discussed in Section 4 and the conclusion of the work is given in Section 5.

2. Proposed dataset

The dataset consists of microscopic blood smear images of ALL, AML, and normal cases. In the dataset, there are 500 microscopic images of blood smears collected from 44 subjects. Images were obtained through a clinical trial from May 2019 to February 2020. The data collection has been done in association with All India Institute of Medical Sciences (AIIMS), Patna, India. The study was conducted after taking approval from institutional ethics committees and as per ethical norms. Informed consent was taken from all the participants and their caretakers. Healthy samples were taken from the Physiology department of AIIMS, Patna, and cancerous samples were taken from the Hematology section of the

Pathology department of AIIMS, Patna. The morphological diagnosis and coordinates of the centroids of probable ALL/AML were given by the Pathology department. Opinion of two experts were taken for labelling a given blood sample as cancerous and also determining the subcategory of acute leukemia. If opinion of both the experts were in agreement then corresponding label was given to the blood sample otherwise it was excluded from the study. All the data were anonymised.

The pathologist prepares the blood smear slides by following the standard method. The blood sample (2 ml) is collected in ethylene diamine tetraacetic acid (EDTA) vacutainer. The smear is produced by putting a drop of blood on a neat and clean slide and spreading by wedge technique. The smear is then allowed to dry. The dried smear is stained with Leishman stain (3 min) followed by Giemsa stain (1 : 10 dilution; 15 min). The slide is washed in running tap water and allowed to dry [22]. The prepared slide is observed under high power (40 $\times$), and oil immersion (100 $\times$) objective lenses.

A Nikon Eclipse Ci microscope coupled to a Nikon DS-Ri2 digital colour camera is used to capture digital images of blood cells present in the blood smear slides. The whole setup is connected to a personal computer. Each image is taken in three-channel, i.e., red-green-blue (RGB) colour space. Images are having colour depth of 8 bits per colour channel, i.e., a total of 24-bit colour depth. Images are acquired at two different magnifications, i.e., 400 $\times$, and 1000 $\times$, corresponding to objective lens 40 $\times$, and 100 $\times$, respectively. The camera is set for automatic exposure and the focus of microscope is adjusted physically by viewing the image on the monitor of the computer. All images are saved in Tagged Image Format File (tiff) with no compression, dimension of 4908 $\times$ 3264 pixels, and resolution of 96 pixels per inch (PPI). The resultant images are raw because normalisation, and colour standardisation techniques are not applied on them.

Table 2

Number of images corresponding to different magnifications and categories.

Magnification	ALL	AML	Normal	Total
400 ×	14	5	26	45
1000 ×	111	120	224	455
Total	125	125	250	500
No. of patients	12	12	20	44

Table 3

Number of ALL images corresponding to different magnifications and sub-categories of ALL.

Magnification	B-ALL	T-ALL	Total
400 ×	10	4	14
1000 ×	93	18	111
Total	103	22	125
No. of patients	10	2	12

Table 4

Number of AML images corresponding to different magnifications and sub-categories of AML.

Magnification	M0	M1	M2	M4	M5	Total
400 ×	1	1	1	1	1	5
1000 ×	21	29	28	20	22	120
Total	22	30	29	21	23	125
No. of patients	2	3	3	2	2	12

The inclusion criterion for cancerous images is that the cases must be of acute leukemia, *i.e.*, ALL, and AML. The exclusion criteria are:

1. subjects suffering from other hematological/hematolymphoid malignancies.
2. Chronic leukemias.

Steps followed to capture an image is as follows: first of all, a region of interest (ROI) is identified by the pathologist looking at the slide under the microscope and then captures image at 400× magnification. Then magnification is increased to 1000× to get more clear and magnified image of blood cells, especially WBC. However, in some cases, images are captured directly at 1000× magnification. On average, 11 images are acquired from a single slide. To get clear images, focus of the camera is adjusted manually.

To date, there are 500 images in the database. Out of 500, 250 are

cancerous, and 250 are normal sample images. The 250 cancerous images contain almost 1700 cancerous blood cells. Samples from the proposed dataset are shown in Fig. 1. In the first row, sample images of normal cases are shown, and in the second and third rows, samples of ALL and AML are shown, respectively. The first column shows the images taken at 400× magnification, and the second and third columns show images taken at 1000× magnification.

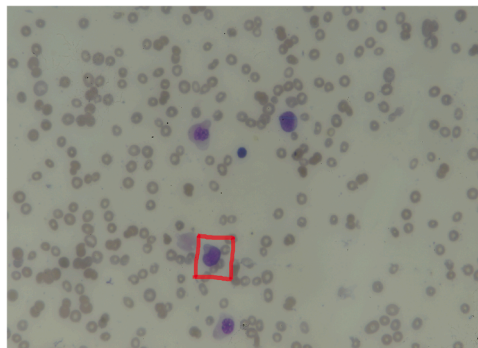
Table 2 summarizes the image distribution. Both ALL and AML are of different types. The dataset currently contains two types of ALL images: B-ALL and T-ALL; and five types of AML images: M0, M1, M2, M4, and M5. Table 3 and Table 4 summarise the image distribution of ALL and AML, respectively.

Fig. 2 displays two images captured at two different magnifications, *i.e.*, Fig. 2a 400× and Fig. 2b 1000 ×, from the same blood slide of AML. Area of interest in 400× image is highlighted with the red rectangle (done manually) by the pathologist and the zoomed image is shown at 1000× magnification.

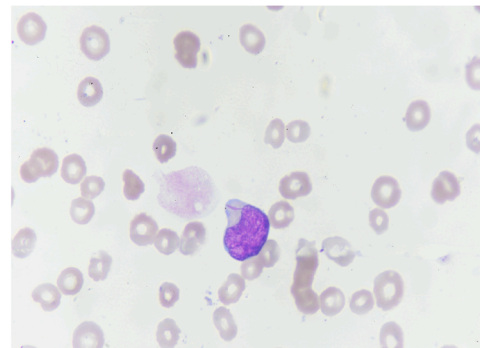
3. Machine learning and deep learning methods for leukemia detection

As discussed in 1.1, several studies have been conducted to identify leukemia. Using digital image processing, features are extracted from images of a dataset and then these features are provided to machine learning algorithms for precise classification. The main hurdle in machine learning technique is the selection of proper feature extractor, as the accuracy of the classifier solely depends on the handcrafted features. Deep learning technique handles this problem very efficiently. Deep learning method extracts relevant information or features automatically, without any human interaction and uses these features efficiently for the classification [23,24].

Our study is broadly divided into three parts. The first two parts are related to binary classification task, *i.e.*, cancerous and normal classes, and the third part consists of three-class classification task, *i.e.*, ALL, AML, and normal classes. In the first part, machine learning approach is used to detect leukemia. Local binary patterns (LBP) [25], and histogram of oriented gradients (HOG) [26] are used as feature extractors and support vector machine (SVM) [27] is used as a classifier. In the second part, transfer learning technique [28] (deep learning approach) is used to detect leukemia. This part is further divided into two sub-parts. In the first sub-part, SVM is trained on two datasets, *i.e.*, the proposed dataset and the mixed dataset. In the second sub-part, a new set of fully connected (FC) layers is used as a classifier. The mixed dataset is used for training and validation purposes. For evaluation of both the sub-parts, testing portion of the proposed dataset and the mixed dataset are used. In the third part, SVM is trained and evaluated on the proposed dataset. In this section, LBP [25], HOG [26], SVM [27], and transfer



(a)



(b)

Fig. 2. Blood slide of AML seen in different magnification factors: (a) 400× and (b) 1000 ×. Area of interest in 400× image is highlighted with the red rectangle (done manually) by the pathologist and the zoomed image is shown at 1000× magnification.

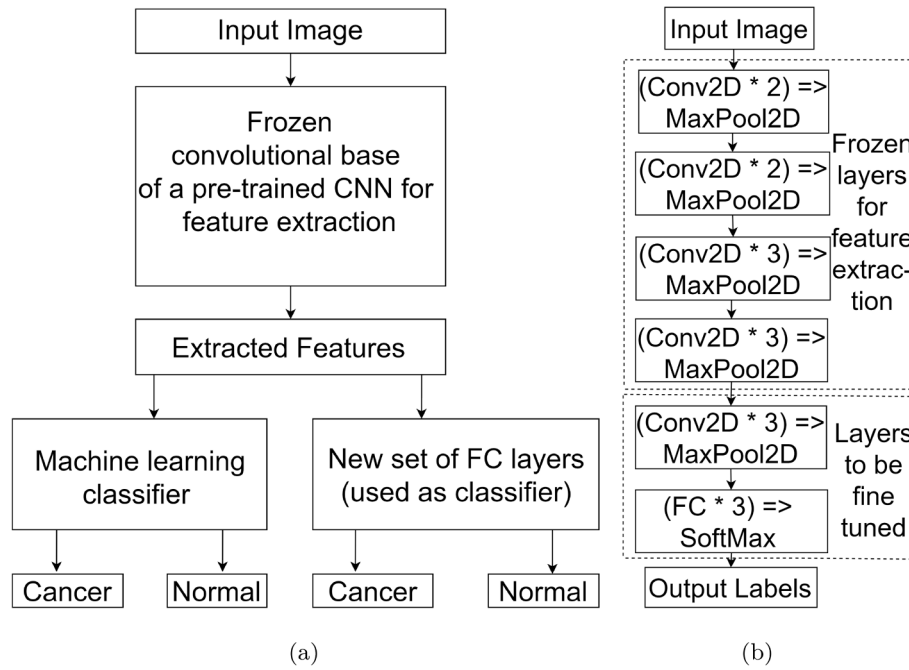


Fig. 3. (a) Overview of the proposed method and (b) fine tuning of FC layers along with the last three convolutional layers of VGG16.

Table 5

Details of data splitting when SVM is used as a classifier in transfer learning technique.

Class	Proposed dataset		Mixed dataset	
	Train	Test	Train	Test
Cancer	200	50	239	60
Normal	200	50	247	62
Total	400	100	486	122

Table 6

Details of data splitting when a new set of FC layers is used as a classifier in transfer learning technique.

Class	Mixed dataset		
	Train	Validation	Test
Cancer	208	31	60
Normal	216	31	62
Total	424	62	122

Table 7

Details of data splitting when SVM is used for three-class classification.

Class	Split 1		Split 2		Split 3	
	Train	Test	Train	Test	Train	Test
ALL	88	37	100	25	113	12
AML	88	37	100	25	113	12
Normal	88	87	100	75	113	62
Total	264	161	300	125	339	86

Table 8

Architectural details of pre-trained CNN models used in the study.

Model	Layers	Parameters (in million)	Input layer size	Output layer size
DenseNet121	121	8	(224, 224, 3)	(2, 1)
InceptionV3	42	24	(299, 299, 3)	(2, 1)
MobileNet	28	4.2	(224, 224, 3)	(2, 1)
ResNet50	50	25.6	(224, 224, 3)	(2, 1)
VGG16	16	138	(224, 224, 3)	(2, 1)
VGG19	19	143	(224, 224, 3)	(2, 1)
Xception	36	22.9	(299, 299, 3)	(2, 1)

learning technique [28] are briefly discussed.

3.1. Local Binary Patterns

LBP [25] is one of the texture descriptors of an image. LBP calculates a mathematical local representation of texture by comparing each pixel of the image with its neighbouring pixels.

The given RGB image is transformed to grey-scale. For each pixel of the transformed grey-scale image, a neighbourhood of radius r having n number of neighbouring pixels is selected. Then a thresholding operation is done with each neighbouring pixel of a central pixel c . If the grey level of one of the neighbouring pixels is greater or same as the grey level of the central pixel then the corresponding neighbouring pixel is assigned 1; otherwise, it is assigned 0. In this way, a binary sequence or pattern is obtained for the central pixel from its neighbourhood. Then the value of the LBP code for the central pixel c is calculated by:

$$LBP(c) = \sum_{k=0}^{n-1} 2^k \cdot d \left(h(q_k) - h(c) \right) \quad (1)$$

where k gives the index value of elements of the binary pattern, $h(q_k)$ and $h(c)$ are intensity values of pixels q_k and c , respectively, and

$$d(y) = \begin{cases} 1 & \text{if } y \geq 0, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

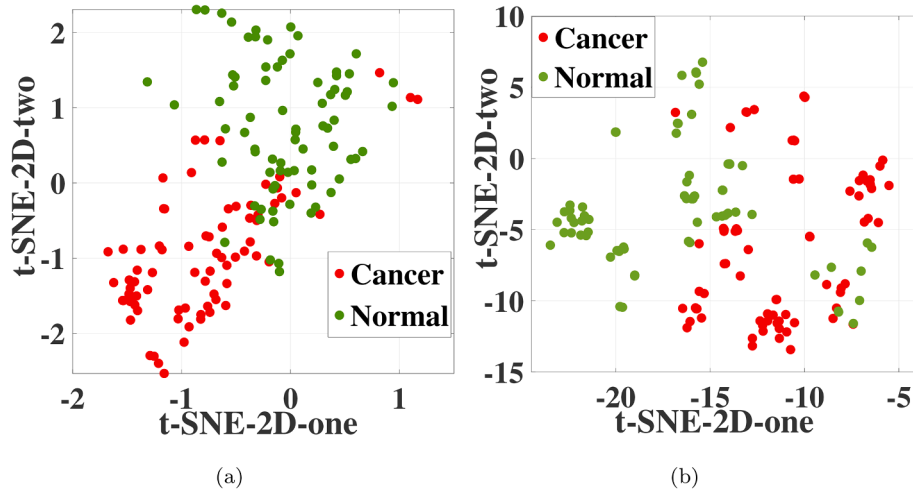


Fig. 4. The t-SNE maps of (a) LBP and (b) HOG.

Table 9

Mean test accuracies of SVM for transfer learning using state-of-the-art CNN models.

Model	Accuracy(%)		
	Proposed dataset	ALL-IDB [16]	Mixed dataset
DenseNet121	89	87	98
InceptionV3	66	56	67
MobileNet	86	86	93
ResNet50	85	89	93
VGG16	89	48	93
VGG19	91	69	93
Xception	77	52	74

This computed LBP code is saved in a 2D matrix of size equal to the input image. These three steps, *i.e.*, thresholding, getting the binary sequence and saving the obtained LBP code in the output 2D matrix, are done for each pixel of the transformed grey-scale image. Histogram of this 2D array is used as a texture descriptor. However, this method has a drawback. Patterns obtained by rotating a binary sequence either in clockwise or anti-clockwise direction and by one step or several steps, posses different LBP codes. Consider a sequence 01000000 and another sequence 10000000, which is obtained by rotating the first one by one

step in the clockwise direction. The LBP codes for the two sequences are 128, and 255, respectively. The remedy of this problem is a rotation invariant LBP code [25]. A given sequence is rotated n number of times and thus, n different LBP codes are obtained. The LBP code having minimum value is considered as the best and final LBP code for the given sequence. After applying rotation invariant LBP method, the LBP code for 01000000, and 10000000 will be the same value. Thus, for a given number of neighbouring pixels around a central pixel, the rotation invariant LBP method reduces the number of possible sequences.

The number of possible patterns can be further reduced with the help

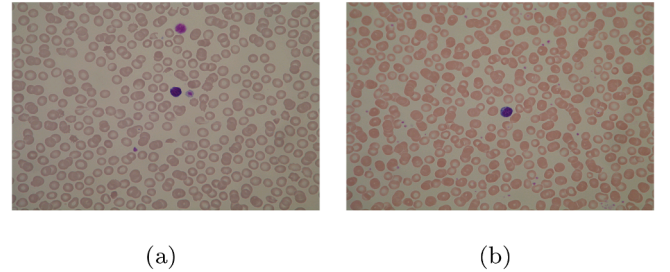


Fig. 6. Samples from normal class misclassified as cancerous when VGG16 + SVM trained on the proposed dataset but correctly classified as normal samples when VGG16 + SVM trained on the mixed dataset.

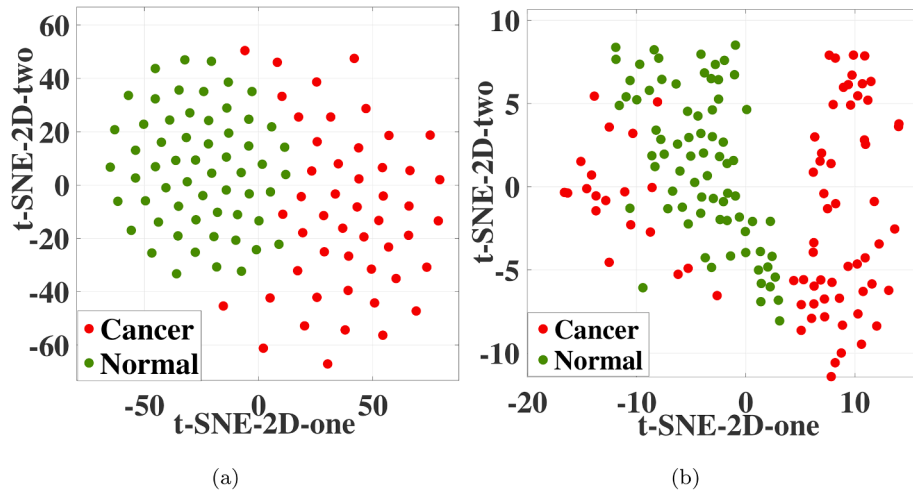
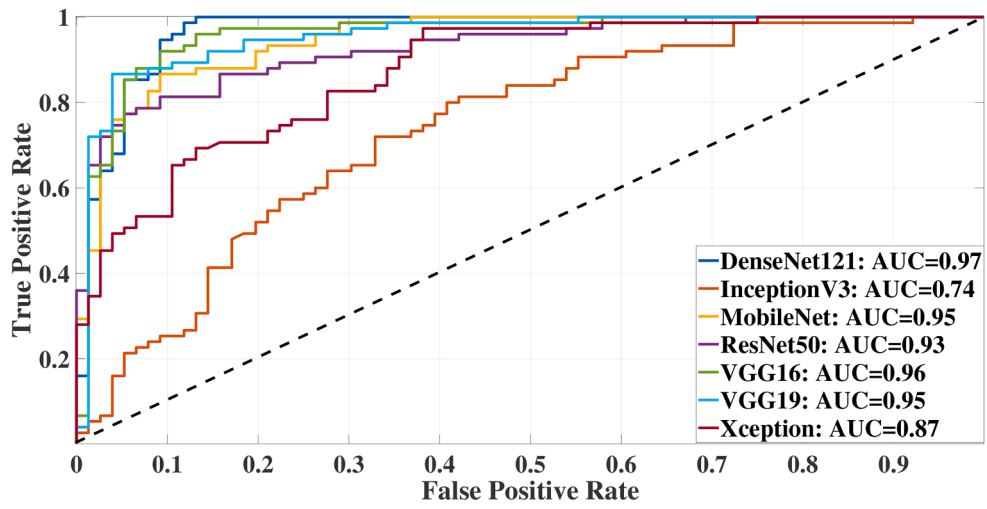
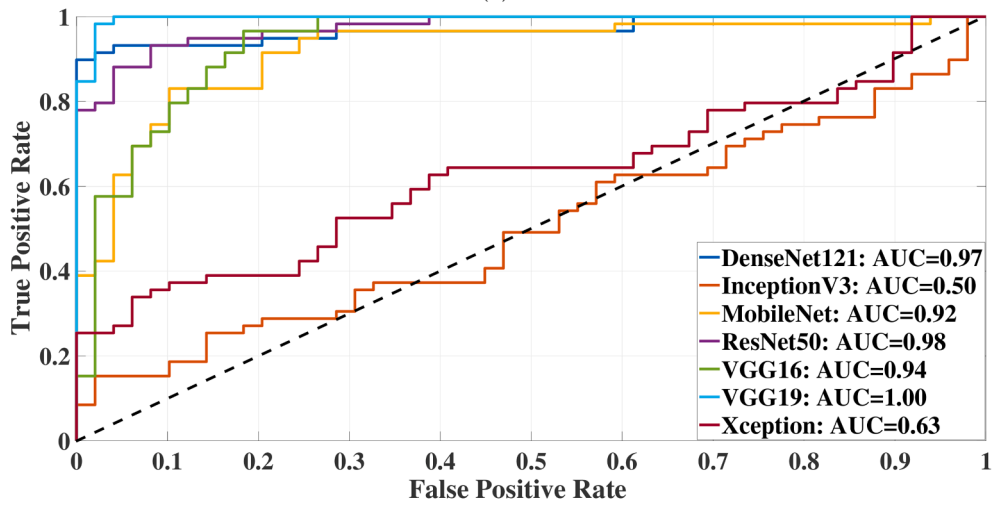


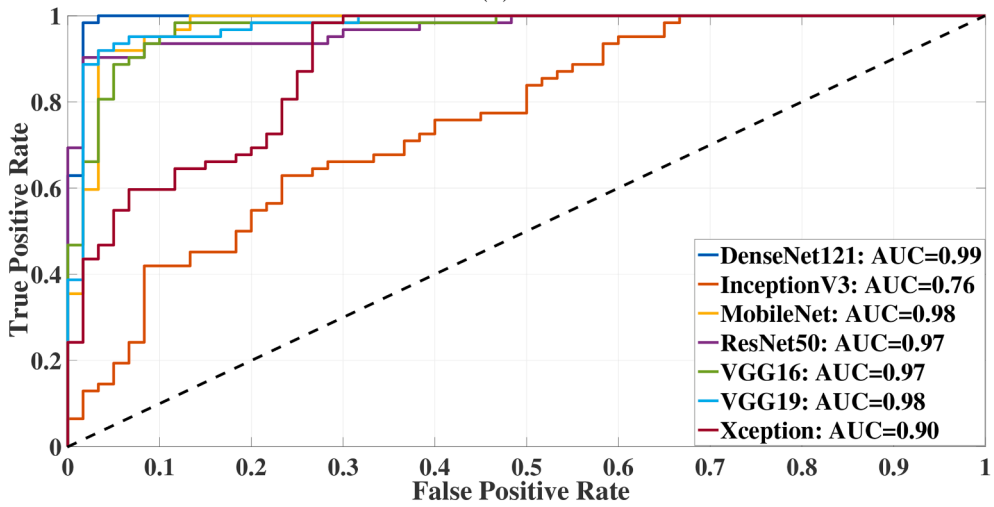
Fig. 5. The t-SNE maps of (a) ALL-IDB [16] and (b) the proposed dataset when VGG16 is used as a feature extractor.



(a)



(b)



(c)

Fig. 7. The ROC curves corresponding to different deep neural networks + SVM used for transfer learning analysis on (a) the proposed dataset, (b) ALL-IDB [16] and (c) the mixed dataset..

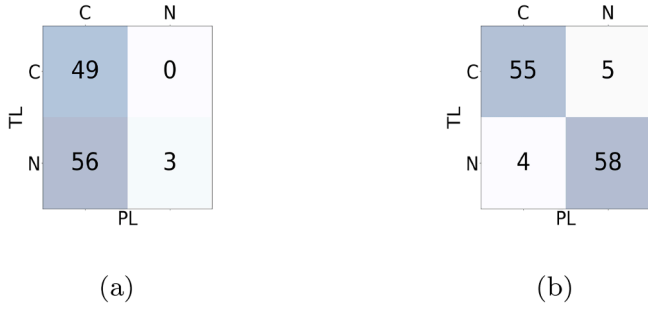


Fig. 8. (a) Confusion matrix obtained when evaluating ALL-IDB [16] on VGG16 + SVM trained on the proposed dataset (b) confusion matrix obtained when evaluating testing part of the mixed dataset on VGG16 + SVM trained on the mixed dataset. TL represents true labels, PL represents predicted labels, C and N stand for cancer and normal labels, respectively.

Table 10

Mean test values of different classification parameters on the proposed dataset after fine tuning CNN models.

Model	Accuracy(%)	Precision	Recall	Specificity	F1 Score
DenseNet121	73	0.73	0.74	0.72	0.73
InceptionV3	49	0.49	0.94	0.04	0.64
MobileNet	59	0.80	0.24	0.94	0.37
ResNet50	55	0.53	0.86	0.24	0.66
VGG16	89	0.85	0.94	0.84	0.89
VGG19	87	0.86	0.88	0.86	0.87
Xception	58	0.55	0.80	0.36	0.65

Table 11

Mean test values of different classification parameters on the mixed dataset after fine tuning CNN models.

Model	Accuracy(%)	Precision	Recall	Specificity	F1 Score
DenseNet121	78	0.78	0.77	0.79	0.77
InceptionV3	49	0.49	0.95	0.05	0.65
MobileNet	64	0.81	0.35	0.92	0.49
ResNet50	60	0.56	0.83	0.37	0.67
VGG16	93	0.90	0.95	0.90	0.92
VGG19	92	0.90	0.93	0.90	0.91
Xception	58	0.56	0.72	0.45	0.63

of uniform patterns [25]. For our analysis, we use rotation-invariant uniform patterns having $n = 24$, and $r = 8$.

3.2. Histogram of Oriented Gradients

HOG [26] are descriptors used for quantifying and representing both shape and texture. The basic principle is that the shape and appearance of an object can be described by the distribution of local intensity gradients, despite not having proper information about the position of gradients. The dimensionality of the final feature vector depends on three factors, in addition to the size of the input image. The factors are: number of *orientations*, *pixels_per_cell*, and *cells_per_block*. An input image is divided into many small rectangular regions known as cells. Two or more cells are combined to form a block. The blocks are formed in such a way that they overlap over one another up to some extent; in other words, each cell contributes to the final feature vector more than once.

Given image is normalised either using gamma law normalisation or square-root equalisation technique. Then image gradient is computed in both x and y directions. The gradient has magnitude as well as angle component (also known as orientation of the gradient). The number of

orientations takes care of the number of bins in the histogram. Next, weighted votes, *i.e.*, gradient magnitude corresponding to each pixel are obtained in spatial and orientation cells. Gradient values are normalised locally to take care of changes in illumination and contrast. For this, a gradient histogram for each cell belonging to a block is computed, concatenated and normalised. Finally, after normalising all the blocks, the resultant histograms corresponding to each block are concatenated and the output is histogram of oriented gradients and treated as the final feature vector.

For our analysis, we use square-root equalisation for image normalisation, number of *orientations* = 9, *pixels_per_cell* = (10, 10), *cells_per_block* = (2, 2), and 'L2-Hys' for block normalisation.

3.3. Support Vector Machines

SVM [27] is a well known classifier which can be helpful for regression and classification tasks, especially when the size of the dataset is small. It is driven by a linear function $\mathcal{W}^T \mathcal{X} + \theta$ which helps in finding an optimal hyperplane for separating two classes [27]. It predicts the class of a new instance $\mathcal{X}$ by simply computing the decision function $\mathcal{W}^T \mathcal{X} + \theta = w_1 x_1 + \dots + w_n x_n + \theta$. If the result is positive, the predicted class $\hat{\phi}$ is the positive class, *i.e.*, 1 and otherwise it is the negative class, *i.e.*, 0.

$$\hat{\phi} = \begin{cases} 0 & \text{if } \mathcal{W}^T \mathcal{X} + \theta < 0, \\ 1 & \text{if } \mathcal{W}^T \mathcal{X} + \theta \geq 0 \end{cases} \quad (3)$$

where θ is the bias term and $\mathcal{W}$ is the feature weights vector.

3.4. Transfer Learning

Transfer learning [28] is the skill to use a pre-trained deep neural network to learn patterns from data on which the neural network was not primarily trained on. The basic idea is to reuse the filters of a pre-trained CNN model for classification task on a dataset which was not originally used to train the CNN model. Basically, transfer learning is applied in two ways:

1. Convolutional base (CNN model without FC layers) of a pre-trained CNN model is used as feature extractor and machine learning model such as Random Forest, Logistic Regression classifier or SVM is used as a classifier. The classifier is trained on the top of features extracted by the convolutional layers to identify new classes of images.
2. The FC layers of a pre-trained network is replaced by a new set of FC layers and the weights of these new layers are fine tuned to identify object classes.

In this work, both the ways of transfer learning are applied to do feature extraction and classification, as shown in Fig. 3a. To further increase performance of a fine tuned CNN model, the immediate neighbouring convolutional layers can also be fine tuned. In our study, we fine tuned FC layers along with the last three convolutional layers of VGG16 [29], as shown in Fig. 3b. We show the fine tuning results in the ensuing section.

4. Experimental Results and Analysis

4.1. Convention

For the first part of the study, *i.e.*, machine learning approach, K -fold cross-validation method [30,31] was used. In our study, $k = 5$ was taken.

In the second part of the study, *i.e.*, using transfer learning technique, the study is divided into two sub-parts. For the first sub-part, training and testing datasets are formed by randomly splitting the proposed

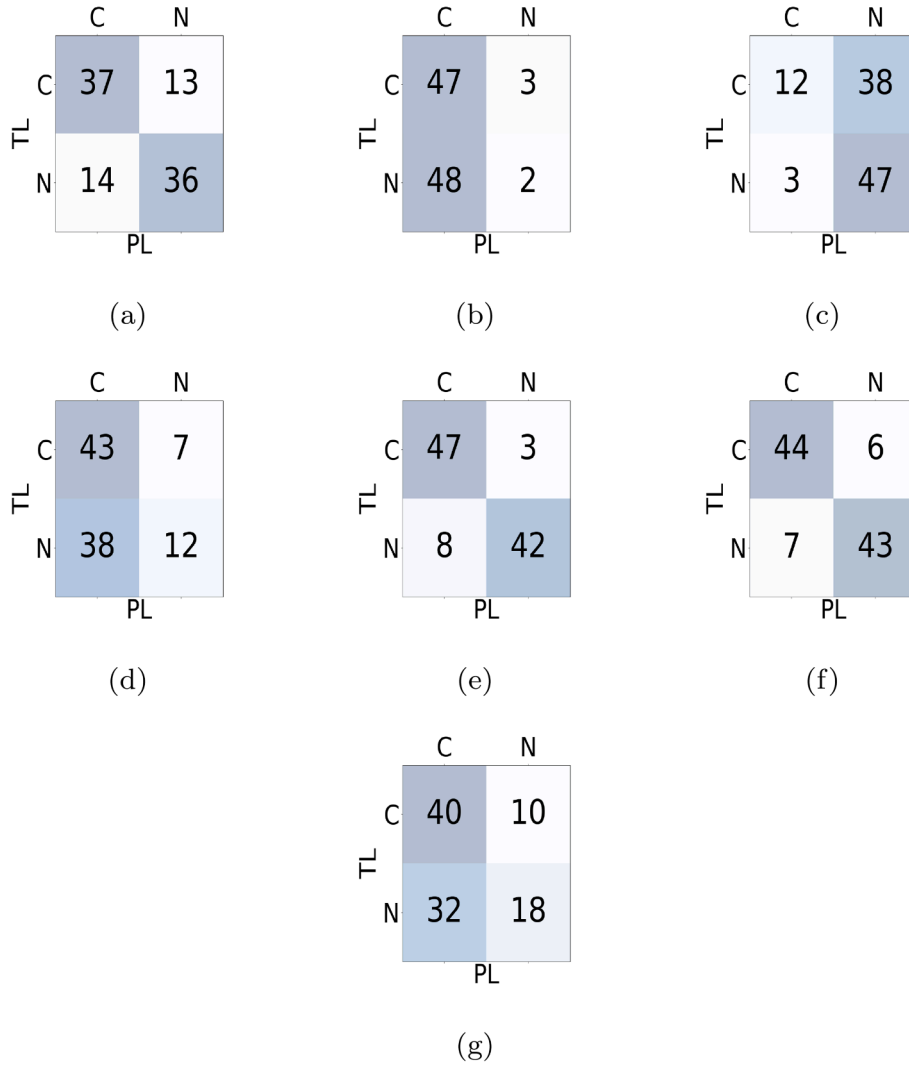


Fig. 9. Confusion matrices produced by fine tuned (a) DenseNet121 (b) InceptionV3 (c) MobileNet (d) ResNet50 (e) VGG16 (f) VGG19 and (g) Xception when evaluated on the proposed dataset. TL represents true labels, PL represents predicted labels, C and N stand for cancer and normal labels, respectively.

dataset and the mixed dataset into two unequal parts, *i.e.*, 80% and 20%, respectively. Table 5 shows data splitting details of this sub-part. Table 6 shows data splitting details of the second sub-part.

For the third part, three splits are done, *i.e.*, split 1, split 2, and split 3. Split 1 contains 264 images for training and 161 images for testing. Split 2 contains 300 images for training and 125 images for testing. Split 3 contains 339 images for training and 86 images for testing. Table 7 shows data splitting details of this part.

For the second and third parts of the study, the input images were initially resized to 224×224 while using DenseNet121 [32], MobileNet [33], ResNet50 [34], VGG16, VGG19 [29], and to 299×299 while using InceptionV3 [35] and Xception [36].

To ensure the classifiers perform well against unseen data, the subjects used in training are not used in testing, *i.e.*, training, validation, and testing sets are subject-independent. Every classifier is trained for five times and for the same number of times it is tested on the testing datasets. Average of these five test performances is taken as the final result.

The SVM was trained using different kernels, however the linear kernel gave the best result and hence, it was retained for further trials. The kernel hyperparameter $C = 0.1$ was empirically determined with the help of the grid search method. Scikit-learn [37], an open-source machine learning library in Python, was used to perform all the experiments related to SVM.

For fine tuning of CNN models, the mixed dataset is used for training,

and testing portion of the proposed dataset and the mixed dataset are used for evaluation. The input images are acquired at different time and condition. Therefore, the images need to be normalized; otherwise, CNN models will not be fine tuned properly [38,39]. Mean subtraction method is used to normalize images, *i.e.*, mean RGB values are subtracted channel-wise from every pixel in the image. After normalizing images they are resized according to the CNN model used. A new set of FC layers is added to the CNN models having output layer size of two for different classes (normal and cancer). Sigmoid activation function is used in the output layer. Different batch sizes, epochs, optimizers and initial learning rates were tried for training; we retained batch size of 10, 20 epochs, stochastic gradient descent (SGD) [40] as optimizer with 0.9 momentum and 0.0001 initial learning rate, which produced the best result. To avoid overfitting, the learning rate was decreased linearly and data augmentation was applied over the training set. After training and saving the models, test accuracy was calculated over the test datasets. The experiments were performed using keras framework [41], an open-source neural network library in Python, and Nvidia RTX 2080 Ti graphics card with dedicated 11 GB memory.

Table 8 shows architectural details of pre-trained CNN models used in the study. DenseNet121 [32] has 120 convolutional layers, four average pooling layers and one fully connected layer. It consists of four dense blocks and between every dense block there is a transition layer. In every dense block there is bottleneck layer that uses 1×1

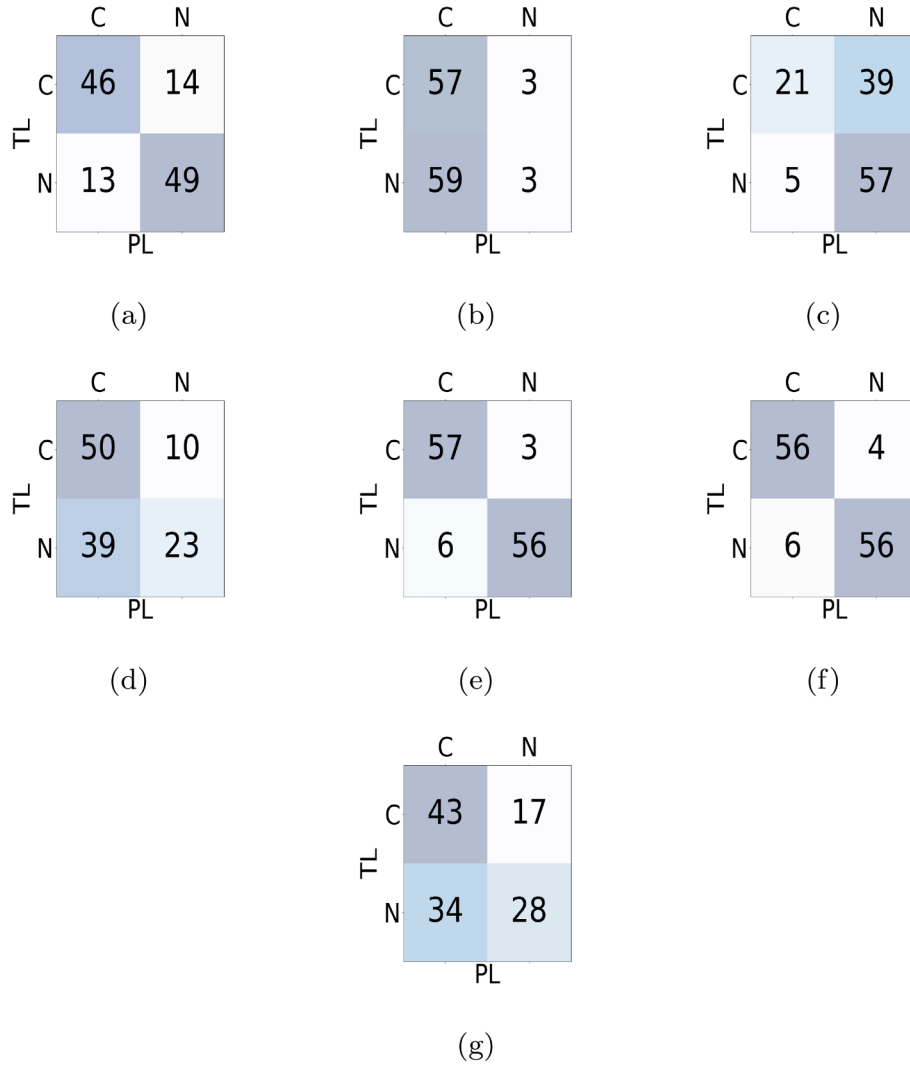


Fig. 10. Confusion matrices produced by fine tuned (a) DenseNet121 (b) InceptionV3 (c) MobileNet (d) ResNet50 (e) VGG16 (f) VGG19 and (g) Xception when evaluated on the mixed dataset. TL represents true labels, PL represents predicted labels, C and N stand for cancer and normal labels, respectively.

convolution. InceptionV3 [35] is a variant of Inception network which consists of 42 layers. It is computational efficient due to use of factorized convolutions. MobileNet [33] consists of 28 layers including all depthwise and pointwise convolutions. Use of depthwise separable convolution helps in reducing model size and complexity. This CNN is mainly used in mobile and embedded vision applications. ResNet50 [34] is a deeper CNN with shortcut connections to handle vanishing gradient problem. In this model, batch normalization (BN) technique [42] is adopted right after each convolution and before activation. Simonyan et al. [29] proposed a deep CNN model known as VGG network. VGGNets use filters of very small receptive field, i.e., 3×3 . VGG16 has 13 convolutional layers and three FC layers whereas VGG19 has 16 convolutional layers. Xception [36] is an extreme version of Inception model but having less number of parameters. This CNN is a linear stack of depthwise separable convolution layers with residual connections and there are total 36 layers in its architecture.

4.2. Results

This subsection contains results of binary classification and three-class classification tasks.

4.2.1. Binary classification

In the first part of the study, SVM is trained with two different de-

scriptors, i.e., LBP and HOG. The test accuracy of SVM trained with LBP is 83% while with HOG is 50%. To assess the difference in performance, the t-distributed stochastic neighbour embedding (t-SNE) method [43] is employed. The t-SNE is a technique which helps in visualising high-dimensional data, e.g. image, in two or three-dimensional map. If in a t-SNE map, data points belonging to different classes can be easily separated from one another, then the performance of feature extractor is better. Thus, t-SNE technique is a useful qualitative indicator. Fig. 4 presents the two-dimensional t-SNE maps of feature extracted by LBP and HOG, which clearly indicates that the performance of LBP is better than HOG.

Table 9 reports the performance of SVM as a classifier when different CNN models are used as feature extractor for transfer learning analysis. SVM is trained on the top of features extracted from our proposed dataset and mixed dataset. The evaluation of trained SVM on our proposed dataset is done on our test dataset and ALL-IDB [16]. Test accuracies corresponding to VGG19, VGG16 [29], and DenseNet121 [32] as feature extractors are 91%, 89%, and 89% respectively. Fig. 5 shows the t-SNE maps of ALL-IDB [16] and the proposed dataset when VGG16 [29] is used as a feature extractor. The t-SNE map corresponding to ALL-IDB shows that the data points belonging to two classes are more clearly separated in comparison to that of the proposed dataset. However, test accuracy corresponding to VGG16 [29] as feature extractor on ALL-IDB [16] reduces to 48%. The difference in performance on two test datasets

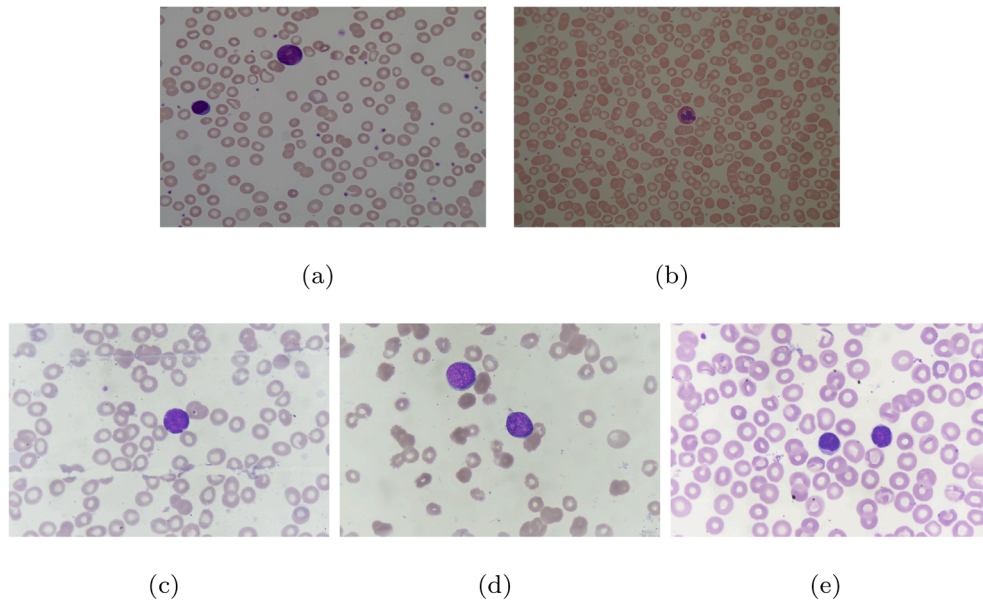


Fig. 11. (a) ALL image and (b) normal blood cells image from ALL-IDB [16] dataset. (c) ALL image (d) AML image and (e) Image containing lymphocyte from the proposed dataset.

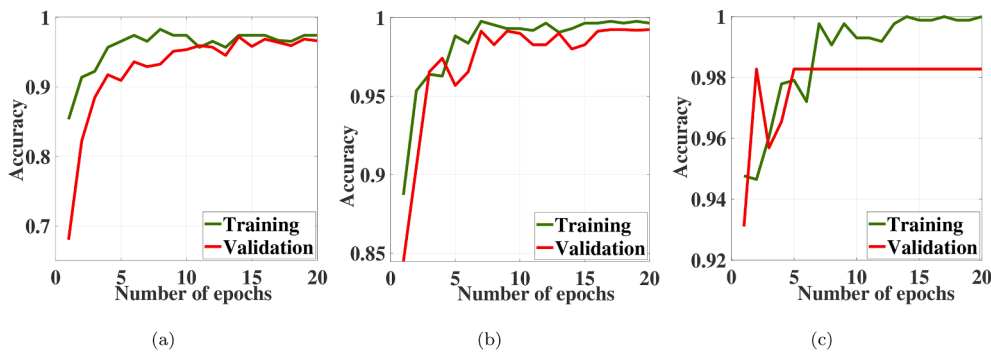


Fig. 12. Curves of training and validation accuracies of fine tuned VGG16 when (a) only FC layers are fine tuned (b) FC + last three convolutional layers are fine tuned and (c) FC + last four convolutional layers are fine tuned.

		C	N
TL	C	47	3
	N	1	49
		PL	

(a)

		C	N
TL	C	57	3
	N	1	61
		PL	

(b)

Fig. 13. Confusion matrices when filters of the last three convolutional layers of VGG16 are fine tuned and evaluated on (a) the proposed dataset and (b) the mixed dataset. TL represents true labels, PL represents predicted labels, C and N stand for cancer and normal labels, respectively.

is observed because images in the two datasets are acquired at different time and condition which lead to severe performance degradation [44]. This difference in performance can be reduced by deeper networks [45,46].

From Table 9 it can be observed that when the SVM is trained on the top of features extracted from the mixed dataset and evaluated on its

testing portion, performance of the SVM corresponding to all CNN models improves on the mixed dataset. SVM trained on the mixed dataset will learn the features of both the datasets (proposed and ALL-IDB [16]) and this leads to improvement in evaluation performance. Fig. 6 shows samples from normal class misclassified as cancerous when VGG16 + SVM is trained on the proposed dataset but correctly classified as normal samples when VGG16 + SVM is trained on the mixed dataset.

Fig. 7 represents receiver operating characteristic (ROC) curves corresponding to different deep neural networks along with SVM used for transfer learning analysis. The performance of a classifier is evaluated with the help of area between the ROC curve corresponding to the classifier and the broken black diagonal line, which represents a random classifier. The area under the ROC curve (AUC) shows a randomly chosen negative sample by a model will have a smaller estimated probability of belonging to the positive class than a randomly chosen positive sample [47].

Fig. 7b shows that AUC corresponding to VGG16 [29] is 0.94 while accuracy is 48%. The contradiction between AUC and accuracy can be understood from Fig. 8a. All the cancerous samples are correctly predicted but normal samples, except 3, are mispredicted as cancerous. However, performance of the SVM improves when it is trained on the top of features extracted from the mixed dataset and evaluated on its testing portion as shown in Fig. 8b.

Tables 10 and 11 show performance of fine tuned DenseNet121 [32],

Table 12

Comparison of the accuracy obtained by the proposed work and other works present in the literature for leukemia detection.

Authors	Classifier used	Dataset used	Accuracy(%)
Madhukar et al. [48] (2012)	SVM	98 images provided by Charles Fabio Scotti	93.50
Agaian et al. [2] (2014)	SVM	ASH [17]	98
Thanh et al. [49] (2018)	CNN	ALL-IDB [16]	96.60
Ahmed et al. [50] (2019)	CNN	ALL-IDB [16]	88.25
Kumar et al. [20] (2020)	CNN	SN-AM Dataset [51]	97.20
Genovese et al. [21] (2021)	VGG16 fine tuned (only last fully connected layers are fine tuned)	ALL-IDB2 [16]	96.84
Renuka et al. [52] (2021)	SVM	ALL-IDB [16]	96.15
Proposed work	VGG16 fine tuned (fully connected layers along with the last three convolutional layers are fine tuned)	Proposed dataset + ALL-IDB [16] (Mixed dataset)	97
	SVM (DenseNet121 along with SVM)		98

Table 13

Test accuracy values when SVM along with state-of-the-art CNN models is used for three class classification.

Model	Accuracy (%)			Average Accuracy (%)
	Split 1	Split 2	Split 3	
DenseNet121	94	94	93	93.67
InceptionV3	76	76	77	76.33
MobileNet	90	94	97	93.67
ResNet50	96	95	99	96.67
VGG16	74	92	97	87.67
VGG19	93	94	95	94.00
Xception	81	81	90	84.00

InceptionV3 [35], MobileNet [33], ResNet50 [34], VGG16, VGG19 [29], and Xception [36] network when they are trained on the mixed dataset and evaluated on the proposed dataset and the mixed dataset respectively. Accuracy, precision, recall, specificity, and F1 score are the classification parameters which are given by

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (4)$$

$$\text{Precision} = TP / (TP + FP) \quad (5)$$

$$\text{Recall} = TP / (TP + FN) \quad (6)$$

$$\text{Specificity} = TN / (TN + FP) \quad (7)$$

$$\text{F1score} = 2 * TP / ((2 * TP) + FP + FN) \quad (8)$$

where TP is true positive meaning that a sample classified as cancerous is actually cancerous. TN is true negative, which implies that a sample predicted as normal is actually normal. FP is false positive, which implies that a sample which belongs to normal category is predicted as cancerous and a sample which belongs to cancerous category however predicted as normal is the case of false negative (FN).

Fig. 9 and Fig. 10 illustrate the confusion matrices produced by fine tuned deep neural networks when evaluated on the proposed dataset

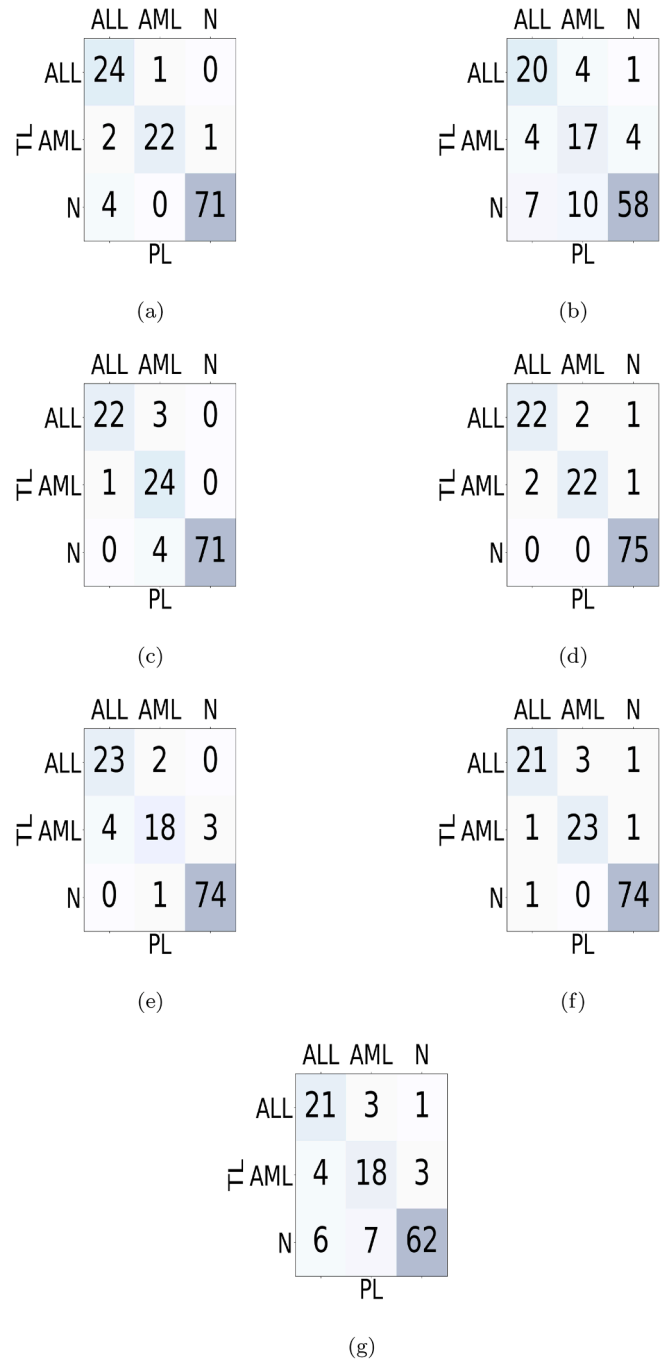


Fig. 14. Confusion matrices produced for SVM as classifier (for Split 2) and (a) DenseNet121 (b) InceptionV3 (c) MobileNet (d) ResNet50 (e) VGG16 (f) VGG19 and (g) Xception used as feature extractor. TL represents true labels, PL represents predicted labels, and N stands for normal label..

and the mixed dataset, respectively. It is evident from the confusion matrices that the performance of almost all fine tuned CNN models, except InceptionV3 [35] and Xception [36] network, improves on the mixed dataset in comparison to the proposed testing dataset.

Tables 10 and 11 show that the performance of fine tuned VGG16 is better than other fine tuned CNN models. Also, test accuracy of fine tuned VGG16 on the mixed dataset is more than that on the proposed test dataset, i.e. 93%, and 89% respectively. The difference in the performance is because the mixed dataset contains some images of ALL-IDB [16] and Fig. 5 shows that the data points belonging to two classes of ALL-IDB [16] are more clearly separable in comparison to that of the

proposed dataset. The proposed dataset contains blasts of two types of acute leukemia and some images of lymphocytes which look morphologically similar to blasts of ALL/AML, although they belong to the normal class; however, ALL-IDB [16] contains only ALL and normal sample images, as shown in Fig. 11.

When the filters of the last three convolutional layers of VGG16 are also fine tuned (Fig. 3b), then the accuracy of 96% on the proposed dataset, and 97% on the mixed dataset are achieved. Fig. 4a and Fig. 4b show curves of training and validation accuracies of fine tuned VGG16 when only FC layers and FC layers along with the last three convolutional layers are fine tuned, respectively. It is evident from these curves that the performance of fine tuned VGG16 improves on fine tuning FC layers along with the last three convolutional layers. On further increasing the number of convolutional layers to be fine tuned, we observe that the model starts to overfit, as shown in Fig. 12c.

Fig. 13 shows the corresponding confusion matrices when the filters of the last three convolutional layers of VGG16 are fine tuned and evaluated on the proposed and the mixed datasets, respectively. The confusion matrices clearly indicate that the performance increases not only for the proposed testing dataset but also for the mixed dataset. Thus, it can be inferred that the proposed dataset enhances the learning ability of the standard CNN models using transfer learning technique.

Table 12 shows a comparison of the accuracy obtained by the proposed work and the existing works presented in the literature. From Table 12, it can be analysed that the proposed method using the combination of DenseNet121 and SVM obtained the accuracy of 98%. However, Agaian et al. [2] used SVM classifier supported by manual feature extraction and got an accuracy of 98%. The total 80 number of sample images were used in this work [2] that is significantly very less number to train an SVM classifier. The proposed work used a heterogeneous dataset consisting of 608 images to train and validate the model. In the proposed work, classification accuracy is improved by fine tuning the last three convolutional layers of VGG16 along with the FC layers which is a novel approach.

4.2.2. Three-class classification

Table 13 shows test accuracy values when SVM along with DenseNet121 [32], InceptionV3 [35], MobileNet [33], ResNet50 [34], VGG16, VGG19 [29], and Xception [36] network is used for three class classification task. It also shows the average accuracy values of three splits for each CNN models. An average accuracy of 96.67% is obtained when SVM is trained on features extracted by ResNet50 [34]. SVM trained on features extracted by VGG19 [29] obtain average accuracy of 94% while for DenseNet121 [32] and MobileNet [33] the average accuracy values are 93.67%.

Fig. 14 illustrates the confusion matrices obtained for SVM as classifier and pre-trained CNN models as feature extractor for split 2. The confusion matrices clearly indicate that the performance of ResNet50 along with SVM is better than other pre-trained CNN models.

5. Conclusion

This work presents acute leukemia binary as well as three-class classification tasks using state-of-the-art machine learning and deep learning techniques on a dataset of microscopic blood smear images. Obtained results are analysed quantitatively in terms of accuracy, precision, recall, specificity, F1 score, and AUC. To analyse the results qualitatively, t-SNE is used. The results of quantitative analysis are in agreement with the qualitative analysis. The binary classification task has achieved an accuracy of 96% on the proposed dataset, and 97% on the mixed dataset where FC layers along with the last three convolutional layers of VGG16 are fine tuned. The new approach is getting better accuracy on the mixed dataset, shows the robustness of the proposed methodology. For three-class classification task, an accuracy of 95% is obtained when SVM is trained on features extracted by ResNet50. Availability of the proposed dataset for the scientific community will

help in developing a computer-assisted diagnosis tool for detecting acute leukemia in the early phase of the disease. Such a tool will prove to be helpful for the pathologists by reducing their burden and decreasing the chances of error. By obtaining the feedback from doctors and medical staff, future studies and new areas of research in this field can be identified. The future work will focus on improving the test performances for both binary and three-class classification tasks.

CRedit authorship contribution statement

Arjun Abhishek: Methodology, Writing - original draft, Writing - review & editing. **Rajib Kumar Jha:** Supervision, Visualization, Investigation. **Ruchi Sinha:** Data curation, Validation. **Kamlesh Jha:** Data curation, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] <https://www.who.int/news-room/fact-sheets/detail/cancer>.
- [2] S. Agaian, M. Madhukar, A.T. Chronopoulos, Automated screening system for acute myelogenous leukemia detection in blood microscopic images, *IEEE Systems Journal* 8 (3) (2014) 995–1004, <https://doi.org/10.1109/JSYST.2014.2308452>.
- [3] E. Pefani, N. Panoskaltis, A. Mantalaris, M.C. Georgiadis, E.N. Pistikopoulos, Chemotherapy drug scheduling for the induction treatment of patients with acute myeloid leukemia, *IEEE Transactions on Biomedical Engineering* 61 (7) (2014) 2049–2056.
- [4] S. Savvopoulos, R. Misener, N. Panoskaltis, E.N. Pistikopoulos, A. Mantalaris, A personalized framework for dynamic modeling of disease trajectories in chronic lymphocytic leukemia, *IEEE Transactions on Biomedical Engineering* 63 (11) (2016) 2396–2404.
- [5] <https://gco.iarc.fr/today/data/factsheets/populations/356-india-fact-sheets.pdf>.
- [6] F. Scotti, Automatic morphological analysis for acute leukemia identification in peripheral blood microscope images, in: CIMS. 2005 IEEE International Conference on Computational Intelligence for Measurement Systems and Applications, 2005, IEEE, 2005, pp. 96–101.
- [7] A.A. Nasir, M. Mashor, H. Rosline, Unsupervised colour segmentation of white blood cell for acute leukaemia images, in: 2011 IEEE International Conference on Imaging Systems and Techniques IEEE, 2011, pp. 142–145.
- [8] D. Mandal, K. Panetta, S. Agaian, Face recognition based on logarithmic local binary patterns, in: *Image Processing: Algorithms and Systems XI*, Vol. 8655, International Society for Optics and Photonics, 2013, p. 865514.
- [9] V.A. Kovalev, A.Y. Grigoriev, H.-S. Ahn, Robust recognition of white blood cell images, in: *Proceedings of 13th International Conference on Pattern Recognition*, Vol. 4, IEEE, 1996, pp. 371–375.
- [10] D.J. Foran, D. Comaniciu, P. Meer, L.A. Goodell, Computer-assisted discrimination among malignant lymphomas and leukemia using immunophenotyping, intelligent image repositories, and telemedicine, *IEEE Transactions on Information Technology in Biomedicine* 4 (4) (2000) 265–273.
- [11] K. Kim, P. Kim, J. Song, Y. Park, Analyzing blood cell image to distinguish its abnormalities, in: *Proceedings of the eighth ACM international conference on Multimedia*, 2000, pp. 395–397.
- [12] J.A. Freeman, D.M. Skapura, *Neural networks - algorithms, applications, and programming techniques*, Computation and neural systems series (1991).
- [13] V. Piuri, F. Scotti, Morphological classification of blood leucocytes by microscope images, in: 2004 IEEE International Conference on Computational Intelligence for Measurement Systems and Applications, 2004. CIMS., IEEE, 2004, pp. 103–108.
- [14] F. Cillesen, W. Der Meer, *Atlas of blood cell differentiation* (1998).
- [15] F. Scotti, Robust segmentation and measurements techniques of white cells in blood microscope images, in: 2006 IEEE Instrumentation and Measurement Technology Conference Proceedings, IEEE, 2006, pp. 43–48.
- [16] R.D. Labati, V. Piuri, F. Scotti, All-idb, The acute lymphoblastic leukemia image database for image processing, in: 18th IEEE International Conference on Image Processing, 2011, pp. 2045–2048.
- [17] A.I. Bank, A. Awards, American society of hematology (2010).
- [18] J. Rawat, A. Singh, H. Bhaduria, J. Virmani, J.S. Devgun, Computer assisted classification framework for prediction of acute lymphoblastic and acute

- myeloblastic leukemia, *Biocybernetics and Biomedical Engineering* 37 (4) (2017) 637–654.
- [19] F. Qin, N. Gao, Y. Peng, Z. Wu, S. Shen, A. Grudtsin, Fine-grained leukocyte classification with deep residual learning for microscopic images, *Computer Methods and Programs in Biomedicine* 162 (2018) 243–252.
 - [20] D. Kumar, N. Jain, A. Khurana, S. Mittal, S.C. Satapathy, R. Senkerik, J. D. Hemanth, Automatic detection of white blood cancer from bone marrow microscopic images using convolutional neural networks, *IEEE Access* 8 (2020) 142521–142531.
 - [21] A. Genovese, M.S. Hosseini, V. Piuri, K.N. Plataniotis, F. Scotti, Acute lymphoblastic leukemia detection based on adaptive unsharping and deep learning, in: *ICASSP 2021–2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2021, pp. 1205–1209.
 - [22] S. Lewis, B. Bain, I. Bates, Dacie and Lewis Practical Haematology, no. pt. 118, p. 2001 in Churchill Livingstone, Churchill Livingstone, 2001. <https://books.google.co.in/books?id=xnZrAAAAMAAJ>.
 - [23] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, *Nature* 521 (7553) (2015) 436–444.
 - [24] A. Pandey, R. Sequeria, P. Kumar, S. Kumar, A multistage deep residual network for biomedical cyber-physical systems, *IEEE Systems Journal* 14 (2) (2019) 1953–1962.
 - [25] T. Ojala, M. Pietikainen, T. Maenpaa, Multiresolution gray-scale and rotation invariant texture classification with local binary patterns, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 24 (7) (2002) 971–987.
 - [26] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection, in: *5 IEEE computer society conference on computer vision and pattern recognition (CVPR'05)*, Vol. 1, IEEE, 2005, pp. 886–893.
 - [27] C. Cortes, V. Vapnik, Support-vector networks, *Machine Learning* 20 (3) (1995) 273–297.
 - [28] S. Bozinovski, Reminder of the first paper on transfer learning in neural networks, 1976, *Informatica* 44 (3) (2020).
 - [29] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, *arXiv preprint arXiv:1409.1556* (2014).
 - [30] R. Kohavi, et al., A study of cross-validation and bootstrap for accuracy estimation and model selection, in: *Ijcai*, Vol. 14, Montreal, Canada, 1995, pp. 1137–1145.
 - [31] J.G. Moreno-Torres, J.A. Sáez, F. Herrera, Study on the impact of partition-induced dataset shift on k-fold cross-validation, *IEEE Transactions on Neural Networks and Learning Systems* 23 (8) (2012) 1304–1312.
 - [32] G. Huang, Z. Liu, L. van der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
 - [33] A.G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, H. Adam, Mobilenets: Efficient convolutional neural networks for mobile vision applications, *arXiv preprint arXiv:1704.04861* (2017).
 - [34] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
 - [35] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2818–2826.
 - [36] F. Chollet, Xception: Deep learning with depthwise separable convolutions, in: *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 1251–1258.
 - [37] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., Scikit-learn: Machine learning in python, *Journal of Machine Learning Research* 12 (Oct) (2011) 2825–2830.
 - [38] S.R. Nayak, D.R. Nayak, U. Sinha, V. Arora, R.B. Pachori, Application of deep learning techniques for detection of covid-19 cases using chest x-ray images: A comprehensive study, *Biomedical Signal Processing and Control* 64 (2021), 102365.
 - [39] Z.N.K. Swati, Q. Zhao, M. Kabir, F. Ali, Z. Ali, S. Ahmed, J. Lu, Brain tumor classification for mr images using transfer learning and fine-tuning, *Computerized Medical Imaging and Graphics* 75 (2019) 34–46.
 - [40] I. Sutskever, J. Martens, G. Dahl, G. Hinton, On the importance of initialization and momentum in deep learning, in: *International conference on machine learning PMLR*, 2013, pp. 1139–1147.
 - [41] F. Chollet, et al., Keras, <https://keras.io> (2015).
 - [42] S. Ioffe, C. Szegedy, Batch normalization: Accelerating deep network training by reducing internal covariate shift, *arXiv preprint arXiv:1502.03167* (2015).
 - [43] L.v.d. Maaten, G. Hinton, Visualizing data using t-sne, *Journal of machine learning research* 9 (Nov) (2008) 2579–2605.
 - [44] P. Pandey, V. Kyatham, D. Mishra, T.R. Dastidar, et al., Target-independent domain adaptation for wbc classification using generative latent search, *IEEE Transactions on Medical Imaging* 39 (12) (2020) 3979–3991.
 - [45] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, T. Darrell, Deep domain confusion: Maximizing for domain invariance, *arXiv preprint arXiv:1412.3474* (2014).
 - [46] X. Glorot, A. Bordes, Y. Bengio, Domain adaptation for large-scale sentiment classification: A deep learning approach, in: *ICML*, 2011.
 - [47] J.A. Hanley, B.J. McNeil, The meaning and use of the area under a receiver operating characteristic (roc) curve, *Radiology* 143 (1) (1982) 29–36.
 - [48] M. Madhukar, S. Agaian, A.T. Chronopoulos, New decision support tool for acute lymphoblastic leukemia classification, in: *Image Processing: Algorithms and Systems X; and Parallel Processing for Imaging Applications II*, Vol. 8295, International Society for Optics and Photonics, 2012, p. 829518.
 - [49] T. Thanh, C. Vununu, S. Atoev, S.-H. Lee, K.-R. Kwon, Leukemia blood cell image classification using convolutional neural network, *International Journal of Computer Theory and Engineering* 10 (2) (2018) 54–58.
 - [50] N. Ahmed, A. Yigit, Z. Isik, A. Alpkocak, Identification of leukemia subtypes from microscopic images using convolutional neural network, *Diagnostics* 9 (3) (2019) 104.
 - [51] <https://doi.org/10.7937/tcia.2019.of2w8l1r>.
 - [52] T.V. Renuka, B. Surekha, Acute-lymphoblastic leukemia detection through deep transfer learning approach of neural network, in: *Proceeding of First Doctoral Symposium on Natural Computing Research: DSNCR 2020*, Vol. 169, Springer Nature, 2021, p. 163.